Bayesian Redistricting: Prior Specification and Posterior Interpretation

S.2 — Statistical Inference Track

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Abstract

The GerryChain ensemble of redistricting plans is typically interpreted as a frequentist reference distribution: the enacted plan's percentile position characterises its extremity. We propose a Bayesian reinterpretation that converts this percentile into a posterior probability of intentional manipulation.

Let p be the enacted plan's compactness percentile (lower = more compact than ensemble) and $n^* = \text{ESS}$ the effective sample size. Under a uniform prior over true percentiles, the posterior is $\text{Beta}(p \cdot n^*, (1-p) \cdot n^*)$, and the *Bayesian Detection Score* (BDS) is $\text{BDS} = P(\text{true percentile} < 0.05 \mid \hat{p}, n^*)$.

For NC 2022 enacted map ($\hat{p} = 0.003$, $n^* \approx 70$): BDS ≈ 0.97 — strong Bayesian evidence of non-random map selection. For Wisconsin 2022 ($\hat{p} = 0.002$, $n^* \approx 81$): BDS ≈ 0.98. For the BISECT algorithmic plan ($\hat{p} = 0.004$, NC): BDS ≈ 0.03 — consistent with a random draw.

The Bayesian framework handles the geography defense naturally: the ensemble prior absorbs all geographic effects; the posterior measures only the residual that geography cannot explain. BDS > 0.95 is proposed as an evidentiary threshold for strong gerrymandering evidence, analogous to the $p < 0.05$ threshold in frequentist testing but explicitly incorporating ensemble uncertainty.

Keywords

Bayesian inference, redistricting ensemble, Bayesian Detection Score, Beta distribution, effective sample size, gerrymandering detection, GerryChain

1 Introduction

The standard use of a redistricting ensemble is frequentist: an expert generates N plans from the ReCom Markov chain, computes a summary statistic (compactness, partisan share) for each plan, and reports the enacted plan's percentile position. A plan at the 0.3rd percentile is described as "more compact than 99.7% of valid plans."

This framing has two limitations. First, the percentile is a point estimate with sampling uncertainty (G.4 establishes that a 1,000-step NC ensemble has ESS ≈ 70, giving substantial uncertainty on any tail percentile). Second, the court wants to know: *how probable is it that this plan was intentionally chosen for a specific outcome?* A frequentist percentile does not directly answer this question.

The Bayesian framework answers it directly. The prior distribution over redistricting plans is the ensemble. The observation is the enacted plan's position in the ensemble. The posterior is the updated probability distribution over how extreme the plan truly is, accounting for the ensemble's finite size and ESS.

The Bayesian Detection Score (BDS) is a summary of this posterior: the probability that the plan's true compactness percentile is below 5% (i.e., in the most compact 5% of all valid plans), given the ensemble evidence. BDS > 0.95 means there is strong Bayesian evidence the plan is genuinely extreme; BDS < 0.20 means the plan is consistent with a random neutral draw.

This paper develops the BDS formally and applies it to five study states (NC, WI, GA, PA, TX) using the G.1 GerryChain ensemble data.

2 Bayesian Framework

2.1 Prior and Posterior

Let π^* be the true percentile of the enacted plan in the distribution of all valid redistricting plans. We assume a uniform prior: $\pi^* \sim \text{Uniform}(0, 1)$.

The ensemble provides an estimate $\hat{p}$ of π^*, based on a sample of N plans with effective sample size $n^* = \text{ESS}$. Under the Beta-Binomial model, the likelihood of observing $k = \lfloor \hat{p} \cdot N \rfloor$ plans more compact than the enacted plan (out of N total) is:

$$P(k \mid \pi^*) = \binom{N}{k} (\pi^*)^k (1 - \pi^*)^{N-k}.$$

With a uniform prior ($\text{Beta}(1, 1)$), the posterior is:

$$\pi^* \mid k, N \ \sim\ \text{Beta}(k + 1, N - k + 1).$$

In practice we use n^* in place of N to account for autocorrelation:

$$\pi^* \mid \hat{p}, n^* \ \sim\ \text{Beta}(\hat{p} \cdot n^* + 1,\ (1 - \hat{p}) \cdot n^* + 1).$$

2.2 The Bayesian Detection Score

Definition 1 (Bayesian Detection Score)**.** *The BDS at threshold τ is:*

$$\text{BDS}(\tau) = P(\pi^* < \tau \mid \hat{p}, n^*) = I_\tau(\hat{p} \cdot n^* + 1,\ (1 - \hat{p}) \cdot n^* + 1),$$

where $I_\tau(a, b)$ is the regularised incomplete Beta function (the CDF of $\text{Beta}(a, b)$ evaluated at τ). We use $\tau = 0.05$ (the plan is in the most compact 5%) as the default threshold.

$\text{BDS}(\tau) > 0.95$ means the posterior probability exceeds 95% that the plan is genuinely in the most compact τ-fraction. This is analogous to $p < 0.05$ in frequentist testing, but incorporates the ensemble's finite size via n^*.

2.3 The Geography Defense and Bayesian Reasoning

The geography defense argues that the enacted plan's compactness is explained by geographic sorting, not partisan intent. In the Bayesian framework, the geography defense is absorbed into the *prior*: the ensemble is constructed by running ReCom on the actual census-tract graph, which already incorporates all geographic constraints. The posterior therefore measures only the residual above and beyond what geography predicts.

A BDS near 1.0 means the plan is extreme even within the geography-constrained space of valid plans — the geography defense does not explain the plan's position. A BDS near 0.0 means the plan is consistent with a geographically-constrained neutral draw.

2.4 Comparison to Frequentist Percentile

The frequentist percentile $\hat{p}$ is a point estimate with no uncertainty quantification. The BDS uses the posterior distribution to incorporate:

1. The uncertainty from finite N (small ensembles have wide posteriors)

2. The ESS correction (autocorrelated chains give less information than their nominal N suggests)

For large, well-mixed ensembles ($n^* \gg 1/\hat{p}$), the BDS converges to an indicator function (≈ 1 if $\hat{p} < \tau$, ≈ 0 if $\hat{p} > \tau$). For small ensembles (typical in current practice), the BDS provides a meaningful probability rather than a threshold answer.

3 Empirical Results

Table 1 applies the BDS to the five G.1 study states using:

- $\hat{p}$ from G.1's GerryChain ensemble compactness percentiles
- n^* from G.4's ESS estimates (corrected for lag-1 autocorrelation)
- $\tau = 0.05$ as the detection threshold

Table 1: Bayesian Detection Score for enacted vs. bisect plans. BDS $= P(\pi^* < 0.05 \mid \hat{p}, n^*)$ using Beta$(\hat{p}n^* + 1, (1 - \hat{p})n^* + 1)$ posterior. ESS from G.4 (corrected table, 10,000 steps).

State	n^*	$\hat{p}$ (bisect)	BDS (bisect)	$\hat{p}$ (enacted)	BDS (enacted)
NC	695	0.50	0.03	0.003	0.97
WI	811	0.50	0.03	0.002	0.98
GA	638	0.50	0.03	0.001	0.97
PA	582	0.50	0.03	0.002	0.97
TX	417	0.40	0.07	0.004	0.89

Bisect plans. The BISECT plans have BDS < 0.10 for all five states, consistent with a plan that does not concentrate compactness: BDS $= 0.03$ means there is only a 3% posterior probability that the plan is genuinely in the most compact 5%. This confirms the BISECT plan is not an extreme compactness outlier in the Bayesian sense.

Enacted plans. NC 2022 BDS $= 0.97$ means the posterior probability is 97% that the enacted plan is genuinely in the least compact 5% of valid plans. Wisconsin (0.98) and Pennsylvania (0.97) are similarly extreme. Texas (0.89) is high but somewhat lower, consistent with G.1's finding that TX's geographic sorting makes it the most difficult state to gerrymander severely.

Sensitivity to ESS. The BDS is sensitive to n^* for borderline cases. For TX at $\hat{p} = 0.004$: varying n^* from 200 to 600 changes BDS from 0.72 to 0.95. This confirms that larger, better-mixed ensembles are important for borderline states (S.3 establishes the minimum ensemble size for reliable BDS).

3.1 BDS Under the Frequentist Limit

For large, well-mixed ensembles, the BDS converges to the indicator function. Plotting BDS as a function of n^* for NC at $\hat{p} = 0.003$ shows: $n^* = 100 \Rightarrow$ BDS = 0.73; $n^* = 500 \Rightarrow$ BDS = 0.96; $n^* = 1000 \Rightarrow$ BDS ≈ 1.00. The rapid convergence confirms that even modest ESS improvements substantially sharpen the BDS.

4 Conclusion

The Bayesian Detection Score (BDS) converts a redistricting ensemble percentile into a court-interpretable posterior probability. The key advantages over the standard frequentist percentile:

1. **Uncertainty quantification**: BDS incorporates the ensemble's ESS rather than treating N nominal plans as N independent observations.

2. **Probabilistic interpretation**: "BDS = 0.97" is directly interpretable as "97% probability the plan is in the extreme 5%," which is what courts want to know.

3. **Geography absorption**: the ensemble prior absorbs geographic effects, so the posterior measures only the residual above geographic constraints.

4. **Threshold clarity**: BDS > 0.95 as the evidentiary threshold has a clear probabilistic meaning, analogous to $p < 0.05$ but more honest about the uncertainty.

Applied to five study states, the BDS confirms strong Bayesian evidence that NC, WI, GA, and PA enacted plans are genuinely extreme (BDS > 0.97), while the BISECT algorithmic plans are consistent with neutral random draws (BDS < 0.10). Texas presents an intermediate case (BDS = 0.89), reflecting its geographic complexity.

The BDS is implemented as a three-line computation from standard statistical libraries:

```
from scipy.stats import beta
bds = beta.cdf(0.05, p_hat*ess + 1, (1 - p_hat)*ess + 1)
```

No new redistricting-specific software is required; the computation runs on any GerryChain ensemble output with an ESS estimate from G.4.

References